

Mind over muscle?

Sex differences in the relationship between body image change and subjective and objective physical changes following a 12-week strength-training program

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Abstract

This study examined sex differences in body image change and its correlates over the course of a 12-week strength-training program. Participants were 28 men and 16 women (M age = 21.6, SD = 2.4) who completed pre- and post-intervention measures of body image (Body Areas Satisfaction Scale, Social Physique Anxiety Scale, and Drive for Muscularity Scale) and subjective and objective assessments of body fat, muscularity, and strength. They participated in a 12-week, 5-day/week full-body progressive resistance training program. Significant body image improvements were found for both sexes ($p < .05$). Correlates of body image change varied somewhat between the sexes. For men, body image improvements were correlated only with subjective physical changes. For women, body image improvements were correlated with subjective physical changes as well as objective increases in strength. These results suggest that although men and women derive body image improvements from strength training, they may benefit for different reasons.

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Introduction

Strength training has been shown to be an effective intervention for enhancing body image among both men and women (for a review, see [Martin & Lichtenberger, 2002](#)). For instance, in a study of male and female college students, a 6-week weight-lifting program significantly improved appearance evaluation, body satisfaction, and social physique anxiety ([Williams & Cash, 2001](#)). Among middle-aged women, a 12-week strength-training program was found to produce

significant improvements in body image that were even larger than those experienced by women in a walking program ([Tucker & Mortell, 1993](#)). And, in a sample of college-aged men, a 4-month strength-training program significantly improved body satisfaction ([Tucker, 1987](#)). Together, these studies highlight the effectiveness of strength training for improving body image in both sexes. However, very little is known about *how* strength training exerts its effects. If the factors that drive body image change can be identified, then strength-training interventions can be designed to be more effective by targeting these mechanisms ([Baranowski, Anderson, & Carmack, 1998](#)).

Several mechanisms have been suggested to explain the effects of strength training on body image. One

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possibility is that body image improves as the result of measurable improvements in strength. Specifically, strength training provides exercisers with feedback regarding their physical capabilities; as exercisers become stronger, they witness an objective increase in their training loads (i.e., the amount of weight they can lift) which, in turn, causes them to feel better about their bodies (Tucker, 1982). Another possibility is that strength training-related decreases in body fat and increased lean body (muscle) mass help people approximate their body ideals, thus improving body image.

Although both of these explanations are tenable, supporting evidence is limited. Some strength-training studies have found that body image improvements are related to strength gains (Tucker & Mortell, 1993) and changes in body composition (e.g., Shaw, Ebbeck, & Snow, 2000; Tucker & Maxwell, 1992), whereas other studies have found no associations between changes in body image, strength (Shaw et al., 2000; Tucker & Maxwell, 1992), and body composition (McAuley, Marquez, Jerome, Blissmer, & Katula, 2002). Furthermore, even when indices of strength and body composition have predicted exercise-related body image change, these indices have accounted for less than 30% of the variance in body image scores. This is probably not surprising given that changes in objective physical characteristics are just one set of variables that can influence body image. Indeed, Cash's (2002) cognitive-behavioral model of body image identifies physical characteristics as one of four developmental influences on body image (in addition to cultural socialization, interpersonal experiences, and personality attributes). Of relevance to understanding the effects of weight training, Cash notes that changes in body weight, muscularity, and physical competence can all influence body image.

Another set of variables that may explain the effects of strength training on body image are changes in people's *perceptions* of their physical characteristics. Strength training can make exercisers feel stronger, thinner, and more toned. These perceived physical changes may elicit improvements in body image independent of objective physical changes (Depcik & Williams, 2004; Martin & Lichtenberger, 2002; McAuley et al., 2002; Sonstroem, 1997). In line with this reasoning, Cash (2002) identifies the internal dialogue of thoughts, interpretations, and conclusions about one's body as one of four sets of proximal events and processes that can affect body image (in addition to appearance-schematic processing, body image emotions, and self-regulatory behaviors).

We are unaware of any published strength-training studies that have examined perceived, subjective

changes in strength and body composition in relation to changes in body image. However, McAuley et al. (2002) tested both subjective and objective measures of physical functioning (self-efficacy and aerobic fitness) and two objective measures of body composition (body fat and fat-free mass) as predictors of body image change among older adults participating in either a "stretch and strength" or an aerobic exercise training program. They found that decreases in social physique anxiety (SPA) were related to subjective and objective changes in physical functioning, but were unrelated to changes in body composition. Thus, McAuley et al.'s results suggest that both subjective and objective physical changes may play a role in body image change associated with various types of exercise. The purpose of the present study was to examine subjective and objective physical changes relative to body image change due to strength training in particular.

We examined the relationship between changes in body image and changes in subjective and objective measures of body fat, muscle, and strength over the course of a 12-week strength-training intervention involving male and female university students. Consistent with McAuley et al.'s findings (2002) and Cash's (2002) model of body image, both subjective and objective physical changes were expected to correlate with changes in body image. However, because body image concerns stem from different factors in young men versus young women, we predicted sex differences in the correlates of body image change.

Specifically, whereas young women's body image concerns stem primarily from concerns about being sufficiently thin (Stice & Whitenton, 2002; Striegel-Moore & Franko, 2002), young men tend to be more concerned with being sufficiently muscular (Cafri & Thompson, 2004; McCabe & Ricciardelli, 2004; McCreary & Sasse, 2000; Olivardia, Pope, Harrison, Borowiecki, & Cohane, 2004).¹ Male body image may also be more strongly predicated on the body's physical capabilities – such as level of strength or endurance – than female body image. For example, based on their

¹ This is not to imply that body weight is of no concern to men, but it is unclear whether male body weight dissatisfaction reflects dissatisfaction with body fat or with lean body mass (i.e., muscle mass). When the two types of dissatisfaction have been measured separately, studies have consistently found muscle dissatisfaction among college-aged men, but findings for body fat dissatisfaction have been inconclusive (see Cafri & Thompson, 2004, for a review). Given the lack of clear evidence for a relationship between body fat and body dissatisfaction in non-obese, college-aged males, we did not predict a relationship between body image change and body fat change.

psychometric analysis of items on the Body Esteem Scale, [Franzoi and Shields \(1984\)](#) reported that women evaluated body parts such as the thighs, waist, and stomach in terms of their physical attractiveness while men evaluated these same parts in terms of their stamina, strength, and agility. Likewise, when asked to rank order a list of body attributes by how important each was to their physical self-concept, men assigned greater importance to attributes based on physical competence (e.g., health and strength), whereas women assigned greater importance to attributes based on physical appearance (e.g., weight and attractiveness; [Fredrickson, Roberts, Noll, Quinn, & Twenge, 1998](#)). Given these sex differences, when men strength train, improvements in their body image may be more related to improved muscularity and strength rather than the development of a thin physique and decreased body fat. Conversely, when women strength train, their body image improvements may be more related to decreases in body fat than increases in strength or muscularity. These hypotheses were tested in the present study.

It was also hypothesized that women would experience greater improvements in body image than men. The rationale for this hypothesis was that women generally report more body image dissatisfaction than men ([Feingold & Mazzella, 1998](#)) and people with the poorest body image tend to experience the greatest body image improvements over the course of exercise interventions ([Martin & Lichtenberger, 2002](#); [Tucker & Maxwell, 1992](#)).

Method

Participants

Participants were 28 men (M age = 21.1 ± 2.3 , range = 18–26) and 16 women (M age = 22.4 ± 2.6 , range = 19–29) who had been sedentary for at least the previous 6 months (i.e., were physically active less than 2 days/week). Approximately 90% of the men and 100% of the women were Caucasian. Participants were recruited from a larger study examining the effects of protein supplementation on physiological changes associated with strength training. During baseline assessments for the larger study, participants were approached by the second author (J.J.E.) and invited to participate in a parallel study of thoughts and feelings about their bodies. It was explained that their involvement entailed completing a packet of questionnaires. Upon completion of the larger study, participants were paid CAN \$500. Sample size was based on the number of

participants needed to detect significant changes in lean body mass determined by dual-energy X-ray absorptiometry (i.e., an outcome of primary relevance to the larger project).

Design

The study utilized a pre-test–post-test quasi-experimental design. All measures were administered at baseline and at the end of the 12-week strength-training program.

Measures

Body image

Consistent with previous studies of strength training and body image ([McAuley et al., 2002](#); [Williams & Cash, 2001](#)), affective and attitudinal dimensions of body image were assessed. *Affect* was measured with the 9-item version of the Social Physique Anxiety Scale (SPAS; [Martin, Rejeski, Leary, McAuley, & Bane, 1997](#)). The SPAS assesses anxiety felt when others observe or evaluate one's physique. Participants indicate the extent to which each scale item is characteristic of them (e.g., "I wish I wasn't so uptight about my physique") using a scale ranging from 1 (*not at all characteristic*) to 5 (*extremely characteristic*). Items are summed such that higher scores indicate greater social physique anxiety. The 9-item SPAS has demonstrated reliability and validity in samples of male and female university students ([Katula, McAuley, Mihalko, & Bane, 1998](#); [Martin et al.](#)). Reliability coefficients were acceptable for both sexes at both measurement points in the present study ($\alpha \geq .81$).

Attitudes were assessed with the 9-item Body Areas Satisfaction Scale (BASS) of the Multidimensional Body-Self Relations Questionnaire (MBSRQ; [Brown, Cash, & Mikulka, 1990](#)). The BASS measures dissatisfaction–satisfaction with nine bodily areas/aspects (e.g., face, upper torso, and muscle tone). Participants rate their level of satisfaction using a scale ranging from 1 (*very dissatisfied*) to 5 (*very satisfied*). Scale items are summed and higher scores indicate greater body satisfaction. The BASS has proven valid and reliable in a previous weight-training study involving male and female college students ([Williams & Cash, 2001](#)). In the present study, reliability was unacceptable for the female sample at baseline ($\alpha = .48$) and post-test ($\alpha = .62$). Deletion of three items with negative item-total correlations (satisfaction with hair, face, and height) increased α to $\geq .74$ for both sexes at

both measurement points.² This 6-item BASS scale was used in all analyses.

The MBSRQ and its subscales have been criticized for not adequately assessing a key dimension of male body image—muscularity satisfaction and concerns (Cafri & Thompson, 2004; McCreary & Sasse, 2000). To address this shortcoming, a male-specific measure of body image attitudes was also administered. The 7-item muscularity attitudes subscale of the Drive for Muscularity Scale (DMS; McCreary & Sasse) assesses attitudes towards one's muscularity. Respondents indicate the extent to which each item applies to them (e.g., “I wish that I were more muscular”) using a scale ranging from 1 (*always*) to 6 (*never*). All items are reverse scored such that higher scores indicate a greater drive for muscularity/muscularity dissatisfaction. The subscale has demonstrated acceptable reliability and validity among college-aged men (McCreary et al., 2004; McCreary & Sasse). McCreary and Sasse caution, however, that this subscale is inappropriate and not psychometrically sound for use with female samples, so data are only reported for the men in our study. The DMS had adequate internal consistency among the men at both study measurement points ($\alpha \geq .89$).

Subjective measures of body fat, muscularity, and strength

Perceived fatness was assessed with the 6-item Body Fat subscale of the Physical Self-Description Questionnaire (PSDQ; Marsh, Richards, Johnson, Roche, & Tremayne, 1994). Respondents indicate how well each item describes them (e.g., “I have too much fat on my body”) using a scale ranging from 1 (*false*) to 6 (*true*). Items are reverse scored and summed so that higher scores indicate less perceived body fat. This subscale has proven valid and reliable in samples of university-aged men and women (Asçi, 2003; Marsh, 1996; Marsh, Hey, Roche, & Perry, 1997; Marsh et al., 1994). In the present study, the scale had adequate internal consistency for both sexes at both measurement points ($\alpha \geq .94$).

Perceived muscularity was assessed using Lynch and Zellner's (1999) male body silhouette scale for the men and Furnham, Titman, and Sleeman's (1994) female body silhouette scale for the women. Participants were shown a series of nine line drawings that ranged in shape from not at all muscular (1) to extremely muscular (9). They were asked to identify the figure “that is most similar to how your body currently looks.” Limited validity data exist for these measures (Furnham et al.; Lynch & Zellner).

Perceived strength was assessed with the 6-item Body Strength subscale of Marsh et al.'s PSDQ (1994). Respondents indicate how well each item describes them (e.g., “I am a physically strong person”) using a scale ranging from 1 (*false*) to 6 (*true*). Items are summed and higher scores indicate greater perceived strength. The Body Strength subscale has proven valid and reliable in samples of university-aged men and women (Asçi, 2003; Marsh, 1996; Marsh et al., 1997, 1994). The scale had adequate internal consistency for both sexes at both measurement points in the present study ($\alpha \geq .91$).

Objective measures of body fat, muscularity, and strength

The objective measures of body fat and muscularity were percentage of body fat (%) and fat- and bone-free mass (kg) as assessed using dual-energy X-ray absorptiometry (DXA: Model QDR-1000/W, Hologic Inc., Waltham, MA). Although fat- and bone-free mass represents more than just muscle, pre- to post-training changes in this variable are considered to be wholly reflective of exercise-induced changes in muscle tissue. Whole body scans were performed from head to toe in the single beam mode, and bone mineral content (BMC), fat, and bone- and fat-free mass were calculated. Participants were scanned with their hands supinated at their sides and their feet placed 20–30 cm apart. Their positions were recorded in order to replicate them at post-testing. In our laboratory, test–retest coefficients of variation for fat and bone free mass on the same individuals are less than .7%. The same investigator performed all of the pre- and post-intervention scans.

The objective measures of strength consisted of voluntary 1-repetition maximum tests (1 RM) for upper- and lower-body muscle groups. 1 RM represents the maximal weight that a person can lift for one repetition and is the “gold standard” for evaluating strength (Kraemer & Fry, 1995). We included the same tests that were used in previous studies of the effects of strength-training on body image (Tucker & Mortell, 1993; Williams & Cash, 2001): the chest press, leg extension, biceps curl, and shoulder press.

² Franzoi and Shields' (1984) factor analysis of the Body Esteem Scale revealed three factors representing: (a) body attributes that cannot generally be changed with exercise, (b) body attributes that can be changed through exercise or control of food intake, and (c) body attributes related to one's physical condition. The three items excluded from the BASS (face, hair, and height) represent attributes that cannot be modified through exercise, whereas the remainder may be (lower torso, upper torso, mid torso, weight, overall appearance, and muscle tone). It is possible that the poor reliability of the BASS in the female sample represents an underlying two-factor scale structure, at least within this sample.

Procedure

After baseline testing, participants began a 5-day/week full-body resistance-training program consisting of exercises performed using guided motion and cable machines. Training was performed at approximately 80% of pre-1 RM for each exercise. All training sessions were monitored by 1 of 14 trainers (upper-year Kinesiology students and certified fitness trainers). A maximum ratio of two participants to one trainer was maintained throughout the study. Study measures were repeated at the end of Week 12.

Results

Preliminary analyses

Three men withdrew from the study before the end of the 12-week training program, resulting in a final male sample of $n = 25$. One woman was unable to perform the Week 12 upper body strength tests because of an injury, but she completed all of the other measures. Thus, the female sample size fluctuates between $n = 15$ and 16 across the analyses. Overall, participants completed 86% of their prescribed exercise sessions (56 sessions prescribed in total).

T-tests indicated that there were no significant differences on the pre- or post-test measures of any

of the study measures between the protein supplement and placebo conditions (all $p > .05$). Accordingly, the data were collapsed across conditions for all subsequent analyses. Exercise adherence was also tested and rejected as a potential covariate for the body image analyses. Computation of Pearson's Product Moment Correlations indicated no significant relationships between adherence and any of the body image measures for either men or women (all $p > .18$).

Changes in body image

Separate 2 (sex) $\times$ 2 (time) general linear model (GLM) repeated measures ANOVAs were conducted on the BASS and SPAS scores. For BASS, significant main effects emerged for time, $F(1, 39) = 35.50$, $p < .001$, partial $\eta^2 = .47$, and for sex, $F(1, 39) = 4.09$, $p < .05$, partial $\eta^2 = .10$. These results indicate that participants experienced significant increases in body satisfaction from pre-intervention ($M = 18.5$, $SD = 4.2$) to post-intervention ($M = 21.9$, $SD = 3.8$), but that overall, women were less satisfied with their bodies ($M = 18.8$, $SD = 3.5$) than men ($M = 21.1$, $SD = 3.5$). Contrary to hypothesis, the time $\times$ gender interaction was not significant, $F(1, 39) = .18$, $p > .05$, partial $\eta^2 = .00$.

Similarly, for SPAS scores, significant main effects emerged for time, $F(1, 39) = 5.21$, $p < .05$, partial

Table 1

Means, standard deviations, significance tests, and effect sizes (partial η^2) for measures of body image and subjective and objective measures of body fat, muscularity, and strength at baseline and Week 12

Measure	Men ($n = 25$)		Women ($n = 16$)		F ($df = 1, 39$) ^a		
	Baseline	Week 12	Baseline	Week 12	Time	Sex	Time $\times$ sex
Body image measures							
Social physique anxiety	23.34 $\pm$ 7.50	21.72 $\pm$ 8.76	30.12 $\pm$ 8.38	28.00 $\pm$ 6.16	5.21* (.12)	7.65** (.16)	.12 (.00)
Body areas satisfaction	23.14 $\pm$ 4.38	26.88 $\pm$ 4.17	21.00 $\pm$ 4.38	24.22 $\pm$ 3.40	35.50* (.47)	4.09* (.10)	.18 (.00)
Muscularity dissatisfaction	17.52 $\pm$ 7.79	14.24 $\pm$ 8.82			14.33* (.37)		
Subjective measures of body fat, muscularity, and strength							
Perceived body fat	28.18 $\pm$ 7.68	28.86 $\pm$ 8.18	21.59 $\pm$ 9.59	24.69 $\pm$ 6.84	4.88* (.11)	4.84* (.11)	2.00 (.05)
Perceived muscularity	4.79 $\pm$ 1.02	5.79 $\pm$ 1.06	4.38 $\pm$.70	5.09 $\pm$.90	43.13* (.53)	1.16 (.10)	3.99 (.03)
Perceived strength	24.44 $\pm$ 6.21	26.92 $\pm$ 5.33	26.81 $\pm$ 5.47	27.69 $\pm$ 5.22	9.39** (.19)	.84 (.02)	2.15 (.05)
Objective measures of body fat, muscularity, and strength							
Body fat (%)	18.22 $\pm$ 5.01	16.38 $\pm$ 4.73	28.61 $\pm$ 7.16	27.96 $\pm$ 6.04	15.40* (.28)	38.61* (.50)	3.52 (.07)
Bone- and fat-free mass (kg)	63.93 $\pm$ 9.48	66.64 $\pm$ 9.05	46.39 $\pm$ 5.59	47.74 $\pm$ 5.53	74.41* (.66)	50.43* (.56)	8.37** (.18)
1 RM shoulder press (kg)	57.63 $\pm$ 13.93	83.80 $\pm$ 22.72	32.47 $\pm$ 6.14	52.60 $\pm$ 6.84	136.57* (.79)	33.95* (.48)	2.32 (.06)
1 RM biceps curl (kg)	47.86 $\pm$ 11.95	68.34 $\pm$ 13.15	21.85 $\pm$ 6.86	37.04 $\pm$ 6.62	181.89* (.83)	76.41* (.67)	3.99 (.10)
1 RM chest press (kg)	72.98 $\pm$ 21.11	100.98 $\pm$ 21.39	37.42 $\pm$ 7.96	62.37 $\pm$ 7.43	394.71* (.91)	44.41* (.54)	1.31 (.03)
1 RM leg press (kg)	99.57 $\pm$ 27.62	170.33 $\pm$ 38.16	67.01 $\pm$ 16.06	109.15 $\pm$ 21.16	234.02 (.86)	31.25* (.45)	15.04* (.28)

Note: Partial η^2 values for each effect are in parentheses.

^a For muscularity attitudes, $df = 1, 24$. For all upper body strength 1 RM measures, $df = 1, 38$.

* $p < .05$.

** $p < .01$.

+ $p < .001$.

$\eta^2 = .12$, and for sex, $F(1, 39) = 7.65$, $p = .01$, partial $\eta^2 = .12$. Social physique anxiety decreased from pre-training ($M = 26.0$, $SD = 8.5$) to post-training ($M = 24.1$, $SD = 8.4$) across the sexes, but women reported greater anxiety ($M = 29.1$, $SD = 7.4$) than men ($M = 22.5$, $SD = 7.4$). Contrary to hypothesis, the time $\times$ gender interaction was not significant, $F(1, 39) = .12$, $p > .05$, partial $\eta^2 = .00$.

A GLM ANOVA on men's DMS scores also revealed effects for time, $F(1, 39) = 14.33$, $p < .001$, partial $\eta^2 = .37$, indicating significant decreases in muscularity dissatisfaction from pre-intervention ($M = 18.5$, $SD = 7.8$) to post-intervention ($M = 13.2$, $SD = 8.8$). Descriptive statistics, F -values, and effect sizes for all variables are presented in Table 1.

Changes in subjective and objective measures of body fat, muscularity, and strength

Prior to examining the relationships between changes in body image and changes in body composition and strength, it seemed prudent to examine whether there were indeed changes in the body fat, muscle, and strength variables. A series of 2 (sex) $\times$ 2 (time) GLM ANOVAs were computed for each measure. As shown in Table 1, significant main effects for time emerged for all of the variables, indicating that participants decreased perceived and objective body fat, and increased their perceived and objective muscle mass and strength over the course of the intervention. In addition, there were significant sex $\times$ time interactions for fat- and bone-free mass and leg strength, indicating that men experienced greater change on these variables than women.

Relationships between changes in body image, body fat, muscularity, and strength

Simple change scores were calculated for each study measure by subtracting the baseline value from the post-test value. For the questionnaire measures, residualized change scores were calculated to control for individual differences in true change and measurement error across the two measurement points (Cohen, Cohen, Aiken, & West, 2003). Residuals were calculated by regressing the post-test score for each measure on the baseline score for that measure. Because individual difference factors do not operate so significantly on physical measures (i.e., correlations between the pre- and post-test objective measures approached 1.00), simple change scores for the objective physical measures can be considered equivalent to residualized change scores (Cohen et al.). One-tailed bivariate correlations were then computed to examine relationships between change in body image and change in body composition and strength. Separate correlations were computed for men and women. The full correlation matrix, showing relationships between all study measures, is presented in Table 2.

The pattern of correlations was somewhat different across the sexes. For men, as expected, increases in body areas satisfaction were associated with increases in perceived strength ($r = .37$) and perceived muscularity ($r = .34$), $p < .05$. Likewise, decreases in social physique anxiety were associated with increases in perceived strength ($r = -.33$), perceived muscularity ($r = -.38$), and, unexpectedly, with changes in perceived body fat ($r = -.47$), $p < .05$. Also unexpectedly, increases in muscularity satisfaction were significantly correlated with perceptions of decreased body fat

Table 2
Bivariate correlations among all study variables for men and women

Changes in	BAS	SPA	DMS	PercBF	PercMUS	PercSTR	BF%	LEAN	Arm	Chest	Shoulder	Leg
Body areas satisfaction (BAS)	–	–.27	–.18	.62**	.46*	.54*	–.23	–.16	.18	.70**	–.06	.46*
Social physique anxiety (SPA)	–.62**	–	–.01	–.43*	–.01	.19	.37	–.22	–.37	–.51*	–.48*	.09
Drive for muscularity (DMS)	.66**	–.47**	–	–.13	.10	–.04	.25	–.32	–.21	–.27	–.04	–.16
Perceived body fat (PercBF)	.26	–.47**	.34*	–	–.03	.21	–.34	.10	.00	.44	.21	.41
Perceived muscularity (PercMUS)	.34*	–.38*	.32	.36*	–	.15	.03	–.51*	–.35	.13	–.18	–.10
Perceived strength (PercSTR)	.37*	–.33*	.13	.23	.20	–	–.10	–.19	–.12	.10	–.31	.84**
Body fat percent (BF%)	.09	.02	–.24	.21	.24	.28	–	–.16	.05	.04	–.21	–.26
Lean mass (LEAN)	.11	.24	.07	–.42*	–.11	–.11	–.34*	–	.05	.20	.12	–.09
Arm strength	–.05	.17	.26	.19	.27	–.01	–.06	.32	–	.53*	.42	.14
Chest strength	–.12	.07	.08	.05	.05	–.24	–.09	–.01	.29	–	.37	.12
Shoulder strength	–.14	.05	–.07	.01	.10	.04	.19	–.25	.13	.69**	–	.06
Leg strength	–.08	.17	–.03	.14	.09	.07	.17	.10	.58**	.31	.18	–

Note: Correlations for the female participants appear above the diagonal, while those for the male participants appear below.

* Correlation is significant at the .05 level (one-tailed).

** Correlation is significant at the .01 level (one-tailed).

($r = .34, p < .05$), but not with increased perceptions of strength ($r = .13, p = .28$). The hypothesized correlation between changes in muscularity satisfaction and perceived muscularity approached, but did not achieve statistical significance ($r = .32, p = .06$). Changes in body image were not associated with changes in any of the objective fat, muscularity, or strength measures.

For women, as hypothesized, improvements in body areas satisfaction and social physique anxiety were correlated with perceived decreases in body fatness ($r = .62$ and $-.43$, for BASS and SPAS, respectively, $p \leq .05$). Yet, contrary to hypothesis, increased body satisfaction was also correlated with increased perceptions of muscularity and strength ($r = .46$ and $.54$, respectively, $p < .05$). No significant correlations emerged between changes in body image and changes in the objective measures of fatness or muscularity. However, increased body areas satisfaction was associated with increased chest ($r = .70$), and leg strength ($r = .46$), $p < .05$. Decreased social physique anxiety was associated with increased shoulder ($r = -.48$) and chest strength ($r = -.51$), $p < .05$.

Discussion

This study examined sex differences in body image change and its correlates over the course of a 12-week strength-training program. Men and women experienced similar significant increases in body areas satisfaction and decreases in social physique anxiety. Men also experienced a significant decrease in muscularity dissatisfaction. However, relationships between body image change and subjective and objective physical changes varied somewhat across the sexes and across the body image measures.

With regard to the male study participants, changes in body image were significantly correlated with subjective, but not objective physical changes. The absence of correlations between body image change and objective physical change is incongruent with Cash's (2002) cognitive-behavioral model of body image. However, different exercisers may interpret the same amount of objective physical improvement very differently, and these interpretations could, in turn, differentially affect other self-perceptions related to body image (Sonstroem, 1997). For example, two men could experience the same 3% decrease in body fat and 3 kg increase in muscle mass, but a man who perceives these physical changes to have greatly improved his body composition would show a greater improvement in body image than a man who perceives very little or no improvement his body composition. Presumably,

subjective responses to objective physical changes would be more closely related to body image change than objective changes per se.

In line with this reasoning, several significant relationships emerged between changes in body image and changes in perceived strength, muscle, and body fat. Interestingly, the pattern of these relationships varied across the three body image measures. With regard to the BASS, as predicted, an increase in body areas satisfaction was associated with an increase in perceived muscularity and strength, but unassociated with changes in perceived body fat. These findings support the notion that beliefs about muscularity (Cafri & Thompson, 2004; McCabe & Ricciardelli, 2004; McCreary & Sasse, 2000) and physical competence (Franzoi & Shields, 1984) are central to men's evaluations of their own bodies, whereas perceptions of body fat are less relevant.

By contrast, changes in perceived body fat – along with muscularity and strength – were correlated with changes in social physique anxiety. Given that SPA reflects self-presentational concerns about the physique (Hart, Leary, & Rejeski, 1989), it seems that whereas perceived body fat is not particularly relevant to a man's evaluations of his own body (i.e., body areas satisfaction), perceived body fat, muscularity, and strength, are all relevant to a man's concerns about *others'* evaluations of his body. Given increasing societal pressure for men to achieve a lean, muscular appearance (Pope, Olivardia, Gruber, & Borowiecki, 1999), it is perhaps unsurprising that self-presentational anxieties decreased as men perceived their bodies approximating societal ideals.

Perceived decreases in body fat were also associated with decreases in muscularity dissatisfaction. This was an unexpected finding, but the relationship likely reflects men becoming more satisfied with the appearance of their muscles as they lost adipose tissue and the underlying musculature became more visible. Also contrary to prediction, changes in muscularity dissatisfaction were not correlated with changes in perceived strength. We interpret these findings as evidence that muscularity dissatisfaction, at least when measured by the DMS, reflects dissatisfaction with muscle appearance and not function.

If muscularity dissatisfaction does indeed reflect appearance-related rather than function-related concerns, then changes in perceived muscularity and muscularity dissatisfaction should be significantly correlated. We found a very modest relationship between these constructs, which did not achieve statistical significance ($p = .06$). On the one hand,

non-significance may reflect insufficient power to detect small relationships. Yet, on the other hand, given that muscularity is such a central feature of male body image (McCreary & Sasse, 2000; Olivardia et al., 2004), perceived muscularity should have been more strongly related to all three body image measures. Perhaps, the body silhouette scale was inappropriate for detecting small changes in perceived muscularity after just 12 weeks of strength training. As an alternative for future strength-training studies, the Somatomorphic Matrix (Gruber, Pope, Borowiecki, & Cohane, 1999) might be a more sensitive index of muscularity perceptions because it provides respondents with a wider range of body silhouettes to choose from.

With regard to correlates of change in women's body image, a somewhat different pattern of relationships emerged. Unlike the male participants, female participants evidenced a significant relationship between objective changes in strength and changes in both body areas satisfaction and SPA. These results were unexpected, but they are consistent with Cash's (2002) body image model, which identifies changes in physical competence as an influence on body image. Furthermore, the results support the notion that strength training provides women with objective positive feedback about their physical capabilities, which, in turn, causes them to feel better about their bodies (Tucker, 1982). Yet, it is not clear why strength gain information would be relevant to women's body image but not men's. Perhaps, men are more influenced by social comparison information (i.e., how strong they are compared to other men), rather than information about their own strength improvements. Another possibility is that because young untrained women are generally weaker than untrained men, women may be more likely to notice and appreciate their objective strength gains during routine activities such as opening heavy doors or carrying groceries. Recognition of strength gains while performing daily activities may also contribute to body image improvements.

As with the men, women showed no significant relationships between changes in body image and changes in objective measures of body fat and muscularity. Again, this speaks to a lack of a direct association between changes in body composition and changes in body image. Body image changes were correlated, however, with subjective changes in body composition. As predicted, decreased SPA was associated with a perceived decrease in body fat, but not with changes in perceived muscularity or strength. These results speak to the centrality of perceived thinness/fatness to a woman's concerns about others'

evaluations of her body. Given that the cultural body ideal for women is a very thin physique, it is understandable that as women perceived their bodies becoming thinner, they had less anxiety about others' evaluations of their bodies.

In contrast, increased body areas satisfaction was correlated with changes in perceived body fat as well as increased strength and muscularity. The latter correlations were unexpected and suggest that women's evaluations of their bodies can be improved by changing perceptions of their strength and muscularity, not just their thinness. Whether these findings are unique to women who choose to strength train remains to be determined. Women who strength train may have stronger, more muscular body image ideals than women who are not motivated to strength train (Furnham et al., 1994).

Limitations and future directions

This study has provided important insights regarding correlates of body image change among novice male and female participants in a strength-training program. Strengths of this study include the parallel assessment of subjective and objective physical changes associated with strength training, and the inclusion of a sex-specific measure of men's body images. Furthermore, the very high level of adherence to the strength-training protocol (no doubt, influenced by the CAN \$500 participant incentive) has provided an excellent test of the efficacy of strength training for improving body image.

Nevertheless, a few limitations warrant mention. First, with only 16 women in the study, the design was underpowered to detect small correlations within the female sample. The study was also underpowered to conduct regression analyses examining the relative contributions of the various correlates to body image change, and, whether perceived physical changes do indeed mediate the effects of objective physical changes. Larger studies are needed to examine these issues.

A second limitation was the concurrent measurement of body image and its correlates; it cannot be concluded that any of the measured variables caused the observed body image changes. However, our identification of sex-specific body image change correlates will inform the design of future prospective studies of mediators of body image change. Future studies should also examine whether perceived physical changes elicit other psychological changes that impact body image. For instance, feelings of goal achievement, satisfaction, and increased self-efficacy derived from perceived physical

improvements could generate a more positive “internal dialogue” (Cash, 2002) about one’s body, which, in turn, could enhance body image.

A third limitation is that verbal interactions between the exercisers and trainers were not controlled. It is possible that exerciser–trainer interactions were qualitatively different across the sexes, and this may have influenced our results. For example, the trainers may have emphasized the effects of strength training on muscularity more with male exercisers than female exercisers, and this may explain why only the men’s SPAS scores were correlated with changes in perceived muscularity.

Conclusion

In summary, our results suggest that although men and women derive similar body image benefits from strength training, they may benefit for somewhat different reasons. Whereas both sexes seem to benefit from strength-training’s effects on *subjective* perceptions of their body composition and strength, women also appear to benefit from experiencing *objective* improvements in strength. Furthermore, changes in different aspects of body image (i.e., body satisfaction, SPA, and muscularity dissatisfaction) are related to different types of physical changes. Prospective studies are now needed to determine whether these changes actually mediate the effects of strength training on body image.

References

- Asçi, F. H. (2003). The effects of physical fitness training on trait anxiety and physical self-concept of female university students. *Psychology of Sport and Exercise*, 4, 255–264.
- Baranowski, T., Anderson, C., & Carmack, C. (1998). Mediating variable framework in physical activity interventions. How are we doing? How might we do better?. *American Journal of Preventive Medicine*, 15, 266–297.
- Brown, T. A., Cash, T. F., & Mikulka, P. J. (1990). Attitudinal body image assessment: Factor analysis of the Body-Self Relations Questionnaire. *Journal of Personality Assessment*, 55, 135–144.
- Cafri, G., & Thompson, J. K. (2004). Measuring male body image: A review of the current methodology. *Psychology of Men and Masculinity*, 5, 18–29.
- Cash, T. (2002). Cognitive-behavioral perspectives on body image. In T. F. Cash, & T. Pruzinsky (Eds.), *Body image: A handbook of theory, research, and clinical practice* (pp. 38–46). New York: Guilford Press.
- Cohen, J., Cohen, P., Aiken, S. G., & West, L. S. (2003). *Applied multiple regression/correlational analysis for the behavioral sciences* (3rd ed.). Mahwah: Lawrence Erlbaum Associates.
- Depcik, E., & Williams, L. (2004). Weight training and body satisfaction of body-image-disturbed college women. *Journal of Applied Sport Psychology*, 16, 287–299.
- Feingold, A., & Mazzella, R. (1998). Gender differences in body image are increasing. *Psychological Science*, 9, 190–195.
- Franzoi, S. L., & Shields, S. A. (1984). The Body Esteem Scale: Multidimensional structure and sex differences in a college population. *Journal of Personality Assessment*, 48, 173–178.
- Fredrickson, B., Roberts, T., Noll, S. M., Quinn, D. M., & Twenge, J. M. (1998). That swimsuit becomes you: Sex differences in self-objectification, restrained eating, and math performance. *Journal of Personality and Social Psychology*, 75, 269–284.
- Furnham, A., Titman, P., & Sleeman, E. (1994). Perception of female body shapes as a function of exercise. *Journal of Social Behavior and Personality*, 9, 335–352.
- Gruber, A., Pope, H. G., Borowiecki, J., & Cohane, G. H. (1999). The development of the somatomorphic matrix: A bi-axial instrument for measuring body image in men and women. In T. S. Olds, J. Dollman, & K. I. Norton (Eds.), *Kinanthropometry VI*. Sydney, Australia: International Society for the Advancement of Kinanthropometry.
- Hart, E. H., Leary, M. R., & Rejeski, W. J. (1989). The measurement of social physique anxiety. *Journal of Sport & Exercise Psychology*, 11, 94–104.
- Katula, J. A., McAuley, E., Mihalko, S. L., & Bane, S. (1998). Mirror, mirror on the wall . . . Exercise environment influences on self-efficacy. *Journal of Social Behavior and Personality*, 13, 319–332.
- Kraemer, W. J., & Fry, A. C. (1995). Strength testing: Development and evaluation of methodology. In P. J. Maud, & C. Foster (Eds.), *Physiological assessment of human fitness* (pp. 115–138). Champaign, IL: Human Kinetics.
- Lynch, S. M., & Zellner, D. A. (1999). Figure preferences in two generations of men: The use of figure drawings illustrating differences in muscle mass. *Sex Roles*, 40, 833–843.
- Marsh, H. W. (1996). Construct validity of Physical Self-Description Questionnaire responses: Relations to external criteria. *Journal of Sport & Exercise Psychology*, 18, 111–131.
- Marsh, H. W., Hey, J., Roche, L. A., & Perry, C. (1997). Structure of physical self-concept: Elite athletes and physical education students. *Journal of Educational Psychology*, 89, 369–380.
- Marsh, H. W., Richards, G. E., Johnson, S., Roche, L., & Tremayne, P. (1994). Physical Self-Description Questionnaire: Psychometric properties and a multitrait-multimethod analysis of relations to existing instruments. *Journal of Sport & Exercise Psychology*, 16, 270–305.
- Martin, K. A., & Lichtenberger, C. M. (2002). Fitness enhancement and body image change. In T. F. Cash, & T. Pruzinsky (Eds.), *Body image: A handbook of theory, research, and clinical practice* (pp. 414–421). New York: Guilford Press.
- Martin, K. A., Rejeski, W. J., Leary, M. R., McAuley, E., & Bane, S. M. (1997). Is the Social Physique Anxiety Scale really multidimensional? Conceptual and statistical arguments for a unidimensional model. *Journal of Sport & Exercise Psychology*, 19, 360–368.
- McAuley, E., Marquez, D. X., Jerome, J., Blissmer, B., & Katula, J. (2002). Physical activity and physique anxiety in older adults: Fitness, and efficacy influences. *Aging and Mental Health*, 6, 222–230.
- McCabe, M. P., & Ricciardelli, L. A. (2004). Body image dissatisfaction among males across the lifespan. A review of past literature. *Journal of Psychosomatic Research*, 56, 675–685.
- McCreary, D. R., & Sasse, D. K. (2000). An exploration of the drive for muscularity in adolescent boys and girls. *Journal of American College Health*, 48, 297–304.

- Olivardia, R., Pope, H. G., Harrison, C., Borowiecki, J., & Cohane, G. H. (2004). Biceps and body image: The relationship between muscularity and self-esteem, depression, and eating disorder symptoms. *Psychology of Men and Masculinity*, 5, 112–120.
- Pope, H. G., Olivardia, R., Gruber, A., & Borowiecki, J. (1999). Evolving ideals of male body image as seen through action toys. *International Journal of Eating Disorders*, 26, 65–72.
- Shaw, J. M., Ebbeck, V., & Snow, C. M. (2000). Body composition and physical self-concept in older women. *Journal of Women & Aging*, 12, 59–75.
- Sonstroem, R. J. (1997). The physical self-system: A mediator of exercise and self-esteem. In K. R. Fox (Ed.), *The physical self: From motivation to well-being* (pp. 3–26). Champaign, IL: Human Kinetics.
- Stice, E., & Whitenton, K. (2002). Risk factors for body dissatisfaction in adolescent girls: A longitudinal investigation. *Developmental Psychology*, 38, 669–678.
- Striegel-Moore, R. H., & Franko, D. L. (2002). Body image issues among girls and women. In T. F. Cash, & T. Pruzinsky (Eds.), *Body image: A handbook of theory, research, and clinical practice* (pp. 183–191). New York: Guilford Press.
- Tucker, L. A. (1982). Effect of a weight training program on the self-concepts of college males. *Perceptual and Motor Skills*, 54, 1055–1061.
- Tucker, L. A. (1987). Effect of weight training on body attitudes: Who benefits most? *Journal of Sports Medicine*, 27, 70–78.
- Tucker, L. A., & Maxwell, K. (1992). Effects of weight training on the emotional well-being and body image of females: Predictors of greatest benefit. *American Journal of Health Promotion*, 6, 338–344.
- Tucker, L. A., & Mortell, R. (1993). Comparison of the effects of walking and weight training programs on body image in middle-aged women: An experimental study. *American Journal of Health Promotion*, 8, 34–42.
- Williams, P. A., & Cash, T. F. (2001). The effects of a circuit weight training program on the body images of college students. *International Journal of Eating Disorders*, 30, 75–82.